

Notes: Martinsson, Sajtos, Strömberg, & Thomann (RFS, 2024)

The Effect of Carbon Pricing on Firm Emissions: Evidence from the Swedish CO₂ Tax

Contents

1	Big picture	3
2	Data and measurement (institutional setting and key objects)	4
2.1	Why Sweden is a useful laboratory	4
2.2	Emissions covered by the tax	4
2.3	Sample and data sources	4
2.4	Main outcome: emission intensity	5
2.5	Main explanatory variable: marginal emissions cost share	5
2.6	Heterogeneity variables	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Short-run event comparisons	6
3.2	Main long-run elasticity regression	6
3.3	Why the sign looks unusual	7
3.4	Industry-year fixed-effect specification	7
3.5	Heterogeneous elasticity specifications	7
3.6	Counterfactual no-tax calculation	7
4	Identification and empirical design (what makes the estimates credible)	8
4.1	Main source of identification	8
4.2	Why marginal cost matters more than average cost	8
4.3	Short-run event evidence as a credibility check	8
4.4	Main threats and how the paper addresses them	9
4.5	What the paper can and cannot show	9
5	Tables and figures (extracted exhibits)	10
5.1	Figure 1. Carbon and energy taxation of an industrial plant	10
5.2	Figure 2. Carbon tax rates over time	10
5.3	Table 1. Swedish carbon tax schedule, part 1	10
5.4	Table 1. Swedish carbon tax schedule, part 2	11

5.5	Table 2. Summary statistics	11
5.6	Figure 3. Changes to the carbon tax by regime	11
5.7	Figure 4. Average and marginal tax rates	11
5.8	Figure 5. Decomposition of manufacturing emissions	13
5.9	Table 3. Short-term tax-regime changes	13
5.10	Table 4. Baseline carbon-pricing elasticities	13
5.11	Table 5. Robustness: EU ETS, firm size, and capital intensity	14
5.12	Table 6. PACE and mobility	14
5.13	Table 7. Carbon leakage classifications	14
5.14	Table 8. Financing constraints	15
5.15	Table 9. Aggregate economic magnitude	15
6	Short summary	16
7	Conclusion	16
7.1	Core conclusion	17
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** Does carbon pricing reduce firm-level carbon emissions, and how large is the response when firms face different marginal carbon costs over a long period?
- **Setting.** Sweden introduced a carbon tax in 1991. The paper studies the universe of Swedish manufacturing firms over 1990–2015, combining firm financial data, heating-related CO₂ emissions, the Swedish tax schedule, tax exemptions, and EU ETS coverage.
- **Why the setting is useful.** Sweden is useful because the tax is large, long-running, and repeatedly changed. Exemptions, rate changes, and EU ETS entry create firm-year variation in the *marginal* cost of emissions, not just one before/after policy date.
- **Core empirical object.** The paper relates changes in firm-level emission intensity, E/Y , to changes in the marginal emissions cost share, C . Here E is firm-level heating-related CO₂ emissions and Y is PPI-adjusted sales.
- **Main baseline result.** A 1% increase in the marginal emissions cost share reduces carbon emissions per unit of PPI-deflated sales by roughly **2% over a three-year period**. In the baseline three-lag specification, the sum of lag coefficients is about **2.08**.
- **Short-run evidence.** Event-style comparisons around the 1991, 1993, and 1997 tax-regime changes show that firms whose marginal tax rises reduce emission intensity relative to firms whose marginal tax falls or remains low.
- **Key heterogeneity result.** Firms in low-abatement-cost sectors respond more strongly. Low-PACE sectors have an elasticity around **3.0**, while high-PACE sectors have an elasticity around **1.9**.
- **Carbon leakage angle.** The highest-emitting sectors and sectors on the EU carbon-leakage list have lower elasticities. This suggests that firms facing high abatement costs or high leakage risk are harder to decarbonize through a uniform carbon price.
- **Financial-friction result.** Less financially constrained firms respond much more strongly to carbon pricing. Public, large, dividend-paying, and mature firms have estimated elasticities mostly between **2 and 3**; more constrained firms often have insignificant elasticities below one.
- **Aggregate magnitude.** Swedish manufacturing heating emissions fell by about **31%** from 1990 to 2015. A simple calibration suggests that 2015 manufacturing emissions intensity would have been about **30% higher** without carbon pricing after accounting for heterogeneity across PACE and mobility groups.
- **Economic takeaway.** Carbon pricing works at the firm level, but its effectiveness is uneven. A high carbon price generates substantial abatement where firms can finance and implement changes; for high-cost or financially constrained firms, complementary policies may be needed.

2 Data and measurement (institutional setting and key objects)

2.1 Why Sweden is a useful laboratory

- Sweden introduced a carbon tax in 1991 and maintained it over a long horizon.
- The statutory tax rate changed many times, and manufacturing firms faced special rules that changed over time.
- The highest emitters were often protected by exemptions based on tax payments as a share of sales; these exemptions lowered their *marginal* tax rates even when their average tax burdens were high.
- The later introduction of the EU ETS created another source of variation because some plants faced permit prices rather than only Swedish carbon tax rates.

Interpretation. The study is not simply comparing Sweden before and after 1991. It exploits many changes in the effective marginal price of emissions across firms and years.

2.2 Emissions covered by the tax

- The paper focuses on stationary manufacturing emissions from fossil-fuel combustion used for heating.
- Process emissions are not the main tax object in the same way; the Swedish carbon tax is levied on fossil fuels used for heating.
- Manufacturing firms must separately declare fossil fuel use for production and heating.
- About two-thirds of the sector's stationary CO₂ emissions are subject to carbon taxation because about one-third of fossil fuel use is tied to production processes.

Interpretation. The tax is targeted at heating-related emissions, so the paper's firm-level outcome is heating-related CO₂ emissions rather than all possible emissions.

2.3 Sample and data sources

- The sample covers Swedish manufacturing firms over **1990–2015**.
- Manufacturing is defined by NACE industries 10–33.
- The main regression sample contains **24,943 firm-year observations** after requiring at least four consecutive observations.
- The paper describes the data set as covering about **4,000 manufacturing firms** and about **85%–90%** of Swedish manufacturing CO₂ emissions during the sample period.

- Firm financial and ownership variables come from UC and Bisnode; emissions data are linked to plant-level information and official emissions reporting.

Interpretation. The sample is broad enough to study aggregate manufacturing emissions but detailed enough to estimate firm-level elasticities and heterogeneity.

2.4 Main outcome: emission intensity

The main firm-level outcome is emission intensity:

$$\frac{E_{it}}{Y_{it}}, \quad (1)$$

where E_{it} is firm i 's heating-related CO₂ emissions in kilograms and Y_{it} is PPI-adjusted sales in constant 2010 Swedish krona.

Interpretation. The paper focuses on emissions per unit of sales rather than total emissions. This emphasizes technique and production changes rather than pure scale changes in output.

2.5 Main explanatory variable: marginal emissions cost share

The key pricing variable is the emissions cost share:

$$C_{it} = \frac{\text{marginal cost of one kg of CO}_2 \times E_{it}}{Y_{it}}. \quad (2)$$

- For tax-only firms, the marginal cost is determined by the Swedish carbon tax schedule and exemptions.
- For EU ETS firms, the marginal cost incorporates the market price of emissions allowances for covered installations.
- The paper uses $\ln(1 - C_{it})$, the log share of sales left after the marginal emissions cost share.

Interpretation. Higher carbon pricing lowers $1 - C$. Therefore, a *positive* coefficient on $\Delta \ln(1 - C)$ means higher carbon pricing reduces emissions intensity.

2.6 Heterogeneity variables

- **Emission-intensity deciles.** Four-digit industries are sorted by 1990 emissions-to-sales ratios into decile groups: D1–D4, D5–D8, and D9–D10.
- **PACE.** Pollution abatement costs and expenditures proxy for the cost of reducing emissions. Low-PACE sectors have lower abatement costs; high-PACE sectors have higher abatement costs.
- **Mobility.** The paper uses real structures capital stock relative to sales to proxy whether production assets are mobile.

- **EU leakage list.** Industries on the EU carbon-leakage list are sectors where carbon pricing is thought to create relocation risk.
- **Financial constraints.** The paper uses public listing, firm size, dividend payout, and firm age as proxies for financing access.

Interpretation. These variables help answer whether carbon pricing is equally effective across firms, or whether abatement costs and financing frictions block the response.

3 Models and empirical specifications (all equations + interpretation)

3.1 Short-run event comparisons

The short-run tests compare firms affected differently by tax changes. A generic two-period difference-in-differences comparison is:

$$\hat{\delta} = \left[\overline{(E/Y)}_{\text{exempt}, \text{post}} - \overline{(E/Y)}_{\text{exempt}, \text{pre}} \right] - \left[\overline{(E/Y)}_{\text{nonexempt}, \text{post}} - \overline{(E/Y)}_{\text{nonexempt}, \text{pre}} \right]. \quad (3)$$

Interpretation. If the exempt group experiences a lower marginal carbon price while the nonexempt group experiences a higher marginal carbon price, a positive $\hat{\delta}$ means the lower-price group becomes relatively more emission intensive.

3.2 Main long-run elasticity regression

The paper's main specification is:

$$\Delta \ln \left(\frac{E_{it}}{Y_{it}} \right) = \alpha + \sum_{s=1}^q \beta_s \Delta \ln(1 - C_{i,t-s}) + \mu_i + \mu_t + \varepsilon_{it}. \quad (4)$$

- E_{it}/Y_{it} is firm-level emissions intensity.
- $C_{i,t-s}$ is the lagged marginal emissions cost share.
- q is the number of lags; the baseline uses $q = 3$.
- μ_i are firm fixed effects.
- μ_t are year fixed effects.

Interpretation. The model asks whether a firm reduces emission intensity when its marginal emissions cost share rises, after removing persistent firm differences and common annual shocks.

3.3 Why the sign looks unusual

Because the right-hand-side variable is $\Delta \ln(1 - C)$, an increase in carbon pricing makes this variable negative. Therefore:

$$\beta_s > 0 \implies \text{higher marginal carbon cost reduces } \Delta \ln(E/Y). \quad (5)$$

Interpretation. Positive coefficients mean carbon pricing works. The paper reports the sum $\sum_s \beta_s$ as the carbon-pricing elasticity.

3.4 Industry-year fixed-effect specification

To address industry-specific time trends and shocks, the paper also estimates:

$$\Delta \ln \left(\frac{E_{it}}{Y_{it}} \right) = \alpha + \sum_{s=1}^q \beta_s \Delta \ln(1 - C_{i,t-s}) + \mu_i + \mu_{j(i),t} + \varepsilon_{it}, \quad (6)$$

where $\mu_{j(i),t}$ is an industry-by-year fixed effect.

Interpretation. This compares firms within the same industry-year, so identification comes from firm-level tax variation generated by exemptions and EU ETS exposure rather than broad industry trends.

3.5 Heterogeneous elasticity specifications

For a subgroup g , the estimated elasticity is:

$$\Delta \ln \left(\frac{E_{it}}{Y_{it}} \right) = \alpha_g + \sum_{s=1}^3 \beta_{g,s} \Delta \ln(1 - C_{i,t-s}) + \mu_i + \mu_t + \varepsilon_{it}. \quad (7)$$

Interpretation. The paper estimates the same carbon-pricing response separately for low/high PACE sectors, low/high mobility sectors, leakage-list sectors, and financially constrained versus unconstrained firms.

3.6 Counterfactual no-tax calculation

The paper uses estimated elasticities to infer what 2015 emissions intensity would have been without carbon pricing:

$$\left(\frac{\widehat{E}_t}{\widehat{Y}_t} \right)^{\text{No tax}} = \left(\frac{E_t}{Y_t} \right) - \left(\sum_{s=1}^3 \widehat{\beta}_s \cdot [\ln(1 - C_{t-s})] \right). \quad (8)$$

Interpretation. The observed emissions intensity is adjusted upward by removing the effect of the carbon prices firms faced in recent years. The exercise is intentionally simple and focuses on intensity, not full general-equilibrium output or relocation effects.

4 Identification and empirical design (what makes the estimates credible)

4.1 Main source of identification

- Identification comes from variation in marginal carbon costs across firms and time.
- This variation is generated by statutory tax-rate changes, tax-payment caps, exemptions for high emitters, special treatment for certain sectors, and EU ETS coverage.
- The paper uses firm fixed effects to remove permanent firm differences.
- It uses year fixed effects, and in some specifications industry-year fixed effects, to absorb broad time shocks.

Credibility logic. The paper is strongest when firms in similar industries and years face different marginal carbon prices because of policy rules, not because of their own contemporaneous emissions decisions alone.

4.2 Why marginal cost matters more than average cost

- A firm's incentive to reduce the next unit of emissions depends on the marginal cost of that unit.
- Under Swedish exemptions, high emitters could pay high average taxes but face low or zero marginal taxes beyond exemption thresholds.
- The paper therefore separates average tax burdens from marginal abatement incentives.

Economic message. High average tax payments do not necessarily generate abatement if the next emitted unit is cheap. The Swedish exemptions are therefore central to the empirical design.

4.3 Short-run event evidence as a credibility check

- The 1991 introduction, 1993 change, and 1997 reintroduction of exemptions create sharp changes in marginal incentives.
- Table 3 shows that groups whose marginal costs rise tend to reduce emission intensity relative to groups whose marginal costs fall or stay low.
- These event results are consistent with the sign of the long-run regression estimates.

Credibility logic. The event evidence shows that the main relationship is visible around specific tax-rule changes, not only in the long panel regression.

4.4 Main threats and how the paper addresses them

- **Common trends.** Year fixed effects and industry-year fixed effects address aggregate or industry-wide trends in technology, demand, or environmental policy.
- **Firm heterogeneity.** Firm fixed effects remove persistent differences in technology, management, and product mix.
- **Concurrent tax reforms.** Robustness tests add corporate-income-tax and labor-tax cost shares.
- **EU ETS effects.** Robustness tests interact carbon pricing with EU ETS coverage.
- **Scale versus technique.** The main outcome is emission intensity, and the aggregate decomposition separates scale, composition, and technique effects.

Interpretation. The design is not a randomized experiment, but the combination of rule-driven variation, fixed effects, event comparisons, and robustness tests makes the carbon-pricing interpretation plausible.

4.5 What the paper can and cannot show

- The paper can estimate firm-level responses of emissions intensity to marginal carbon prices.
- It can show heterogeneity by abatement costs, mobility, leakage risk, and financial constraints.
- It cannot fully quantify general-equilibrium effects, such as output relocation, downstream price changes, or global leakage.
- The counterfactual is a back-of-the-envelope calibration rather than a structural welfare calculation.

Takeaway. The cleanest contribution is a long-panel estimate of carbon-pricing elasticities. The aggregate calibration is useful for magnitude but should be interpreted as an intensity-based exercise.

5 Tables and figures (extracted exhibits)

5.1 Figure 1. Carbon and energy taxation of an industrial plant

Read: The figure separates heating emissions, process emissions, utility emissions, and mobile emissions. The Swedish manufacturing carbon tax mainly targets heating-related fossil-fuel combustion.

Economic message. The outcome variable is not generic firm emissions. The paper studies the part of manufacturing emissions where the Swedish carbon tax directly changes marginal incentives.

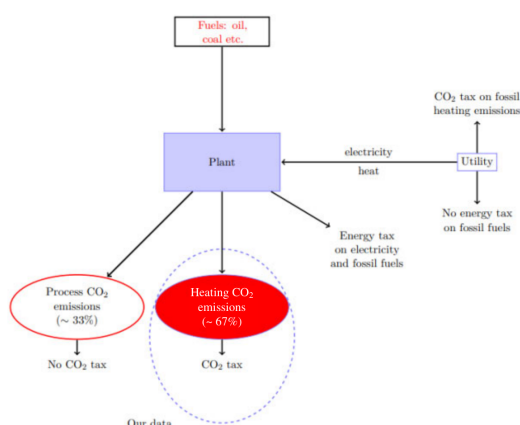


Figure 1
Carbon and energy taxation of an industrial plant
This figure illustrates the carbon and energy taxation for a manufacturing plant in Sweden in 2019. *Heating CO₂ emissions* refers to the emissions released from the combustion of fossil fuels. *Process CO₂ emissions* refers to the carbon dioxide emissions released in the actual manufacturing process (i.e., not combustion of fossil fuels). *Utility* is the power plant that produces heat and/or electricity, and *Plant* is the industrial manufacturing plant.

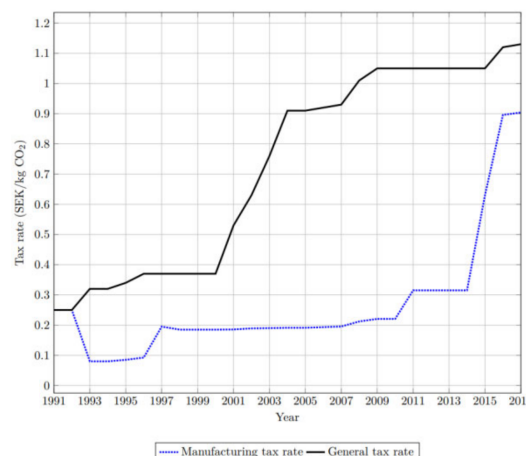


Figure 2
Carbon tax rate, in nominal values
This figure displays the nominal carbon tax rates (Swedish krona per kilogram of emitted carbon dioxide) for Sweden from 1991 to 2017. *Manufacturing tax rate* refers to the tax rate for the manufacturing sector (SNI 10-33 in the SNI2007 nomenclature), while *General tax rate* refers to the tax rate for nonindustrial firms and households.

5.2 Figure 2. Carbon tax rates over time

Read: The standard carbon tax rate rises sharply over time, while the manufacturing rate is lower for much of the sample because of special treatment and exemptions.

Economic message. Sweden provides a high and changing carbon price. The gap between general and manufacturing rates explains why firm-level effective marginal prices are more informative than the headline tax rate.

5.3 Table 1. Swedish carbon tax schedule, part 1

Read: Table 1 summarizes the changing rules, including standard rates, manufacturing rates, general exemptions, special rules for cement/glass/lime, and EU ETS treatment.

Economic message. The table is the institutional backbone of the paper. The empirical variation comes from changing tax rates and exemptions, not merely from the initial adoption of the tax.

Table 1
Summary of the rates in the Swedish carbon tax system

Year	Carbon tax rates (SEK/kg)					Firms in EU ETS
	Standard rate	Manufacturing rate	General exemptions	Cement, glass lime		
1990	No tax	No tax	No tax	No tax		
1991	0.25	0.25	Manufacturing rates if CO ₂ + Energy tax <= 1.7% of sale, untaxed further emissions	Manufacturing rates if CO ₂ + Energy tax <= 1.7% of sale, untaxed further emissions		
1992	0.25	0.25	Manufacturing rates if CO ₂ + Energy tax <= 1.2% of sale, untaxed further emissions	Manufacturing rates if CO ₂ + Energy tax <= 1.2% of sale, untaxed further emissions		Before EU ETS
1993	0.32	0.08	Manufacturing rate	Industry rate up to 1.2 % of sales, untaxed further emissions ("1.2% rule")		
1994	0.32	0.08				
1995	0.34	0.09				
1996	0.37	0.09				
1997	0.37	0.19				
1998	0.37	0.19				
1999	0.37	0.19				
2000	0.37	0.19				
2001	0.53	0.19				
2002	0.63	0.19				
2003	0.76	0.19				
2004	0.91	0.19				
2005	0.91	0.19	Manufacturing tax rate up to 0.8% of sales, exceeding emissions: 25 % of general manufacturing CO ₂ tax rate ("0.8 % rule")			
2006	0.92	0.19			Manufacturing rate + exemptions where applicable	

(Continued)

Table 1
(Continued)

Year	Carbon tax rates (SEK/kg)					Firms in EU ETS
	Standard rate	Manufacturing rate	General exemptions	Cement, glass lime		
2007	0.93	0.20			Special exemption removed	
2008	1.01	0.21				
2009	1.05	0.22				EU ETS+15% of standard rate for plants under EU ETS
2010	1.05	0.22				
2011	1.05	0.315				
2012	1.05	0.32				
2013	1.05	0.32	Manufacturing rate up to 1.2% (Exceeding: 24% of manufacturing rate)			No CO ₂ tax for installations covered by EU ETS
2014	1.05	0.32				
2015	1.05	0.63	Special exemption removed			

This table summarizes the special provisions that exempt tax reliefs for certain industrial enterprises. Standard rate applies for households and nonindustrial firms. Manufacturing rate is the applicable rate for manufacturing enterprises (SNI10-33 under SNI2007 nomenclature), and the exemptions in Manufacturing rate + exemptions where applicable are the 0.8% and the 1.2% rules.

5.4 Table 1. Swedish carbon tax schedule, part 2

Read: The continuation of Table 1 shows the 2007–2015 period, including removal of special exemptions and the interaction with EU ETS coverage.

Economic message. Once EU ETS becomes important, marginal incentives for high emitters change substantially. This is one reason the post-2005 calibration implies large abatement effects.

5.5 Table 2. Summary statistics

Read: The main sample has 24,943 observations. Mean emissions-to-sales is 0.0072, and the mean marginal emissions-cost-to-sales variable is 0.0010. The table also reports EU ETS status, PACE, mobility, leakage-list status, and financing proxies.

Economic message. The data allow the paper to connect emissions, carbon pricing, firm size, capital intensity, sector classifications, and ownership/financing variables in the same panel.

5.6 Figure 3. Changes to the carbon tax by regime

Read: The figure shows how a representative firm's carbon tax payments differ across regimes. The exemption rules cause marginal and average tax rates to diverge.

Economic message. Exemptions can protect firms from high marginal tax rates even when they pay substantial average taxes. This distinction is central for interpreting firm behavior.

5.7 Figure 4. Average and marginal tax rates

Read: Nonexempt, non-EU-ETS firms face similar average and marginal tax rates. Exempt/EU-ETS firms often face very different average and marginal incentives.

Economic message. The paper estimates responses to the marginal price, because that is the price relevant for an additional unit of emissions.

Table 2
Summary statistics

	OBS	Mean	Median	SD	Min	Max
<i>Emissions-to-sales</i>	24,943	0.0072	0.0021	0.0184	0.0000	> 0.100
<i>MC of emissions-to-sales</i>	24,943	0.0010	0.0004	0.0017	0.0000	0.0104
<i>MC of emissions-to-EBIT</i>	24,904	0.0143	0.0034	0.0995	-0.4164	0.5900
<i>Nr of workers</i>	24,884	234	45	908	0	n/a
<i>Capital intensity</i>	24,682	0.6083	0.3255	0.9010	0.0022	5.6367
<i>EU ETS</i>	24,943	0.0673	0.0000	0.2505	0.0000	1.0000
<i>Low pace</i>	24,943	0.4658	1.0000	0.4988	0.0000	1.0000
<i>High pace</i>	24,943	0.4788	0.0000	0.4996	0.0000	1.0000
<i>Low pace & Low mobility</i>	24,943	0.2064	0.0000	0.4047	0.0000	1.0000
<i>Low pace & High mobility</i>	24,943	0.2547	0.0000	0.4357	0.0000	1.0000
<i>High pace & Low mobility</i>	24,943	0.2927	0.0000	0.4550	0.0000	1.0000
<i>High pace & High mobility</i>	24,943	0.1780	0.0000	0.3825	0.0000	1.0000
<i>Not on leakage list</i>	24,943	0.5043	1.0000	0.5000	0.0000	1.0000
<i>On leakage list</i>	24,943	0.4931	0.0000	0.5000	0.0000	1.0000
<i>D1-D4</i>	24,943	0.3969	0.0000	0.4893	0.0000	1.0000
<i>D5-D8</i>	24,943	0.4265	0.0000	0.4946	0.0000	1.0000
<i>D9-D10</i>	24,943	0.1725	0.0000	0.3778	0.0000	1.0000
<i>Public firm</i>	16,328	0.2257	0.0000	0.4181	0.0000	1.0000
<i>Private firm</i>	16,328	0.7743	1.0000	0.4181	0.0000	1.0000
<i>Large firm</i>	16,328	0.4789	0.0000	0.4996	0.0000	1.0000
<i>Small firm</i>	16,328	0.5211	1.0000	0.4996	0.0000	1.0000
<i>High dividend firm</i>	16,328	0.5137	1.0000	0.4998	0.0000	1.0000
<i>Low dividend firm</i>	16,328	0.4863	0.0000	0.4998	0.0000	1.0000
<i>Mature firm</i>	16,328	0.4995	0.0000	0.5000	0.0000	1.0000
<i>Young firm</i>	16,328	0.5005	1.0000	0.5000	0.0000	1.0000

This table reports summary statistics in the key variables included in this study. The firm-level data are from UC and Bisnode and consist of CO₂-emitting firms with at least four consecutive observations over 1990–2015 and a primary NACE industry classification between 10 and 33. Monetary values are adjusted and expressed in constant 2010 Swedish krona (SEK). CO₂ emissions are expressed in kilograms (kg). *MC of emissions-to-sales* is the emissions cost (marginal cost multiplied by emissions) share relative to sales for firm *i* in year *t*. *MC of emissions-to-EBIT* is relative to earnings before interest and taxes. *Capital intensity* is the ratio between fixed assets and workers. *EU ETS* is an indicator variable taking on the value one if the firm is regulated under EU ETS some time during the sample period, and zero otherwise. *Low (High) pace* (pollution abatement costs and expenditure) is an indicator variable taking on the value one if the firm is located in an industry below (above) the median in terms of air pollution abatement costs and expenditures relative to sales, and zero otherwise. *Low (High) mobility* is an indicator variable taking on the value one if the firm is located in an industry above (below) the median in terms of the real structures capital stock to sales, and zero otherwise. *Not on leakage list* and *On leakage list* are indicator variables taking on the value one if the sector the firm is operating in is either not on or on the EU's Carbon leakage list. *D1-D4* is an indicator variable taking on the value one if the firm is located in an industry in the first to fourth decile in terms of CO₂ emission to sales in 1990, and zero otherwise. *D5-D8* and *D9-D10* are based on firms from deciles 5–8 and 9–10, respectively. Decile 1 (10) means lowest (highest) emission intensity. The firm ownership and financing variables are available over 1996–2015. *Public (Private) firm* is an indicator variable taking on the value one (zero) if the firm is (not) listed on a Swedish stock exchange. A firm is considered publicly listed if at least one firm in the corporate group is publicly listed at least once during the sample period. *Large (Small) firm* is an indicator variable taking on the value one (zero) if the firm is above (below) the median in book value of total assets (averaged over the sample period and measured at the corporate group level) within its four-digit NACE industry. *High (Low) dividend firm* is an indicator variable taking on the value one (zero) if the firm is above (below) the median in dividend payout divided by book value of total assets (averaged over the sample period and measured at the corporate group level) within its four-digit NACE industry. *Mature (Young) firm* is an indicator variable taking on the value one (zero) if the firm's founding year (measured at the corporate group level) is below (above) the median within its four-digit NACE industry.

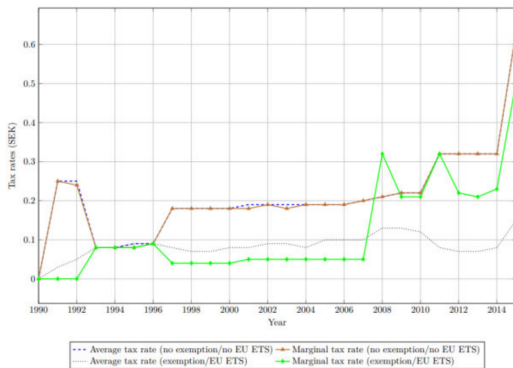


Figure 4
Average and marginal tax rates (1990–2015)

This figure displays the average and marginal tax rates depending on whether the firm is eligible for carbon tax exemptions and covered by the EU ETS. *no exemption/no EU ETS* denotes firms that are not regulated by the EU ETS and are not entitled to carbon tax exemption, *exemption/EU ETS* refers to the firms with available exemptions until they enter the emission trading scheme. Marginal tax rates for EU ETS are the price for emission rights.

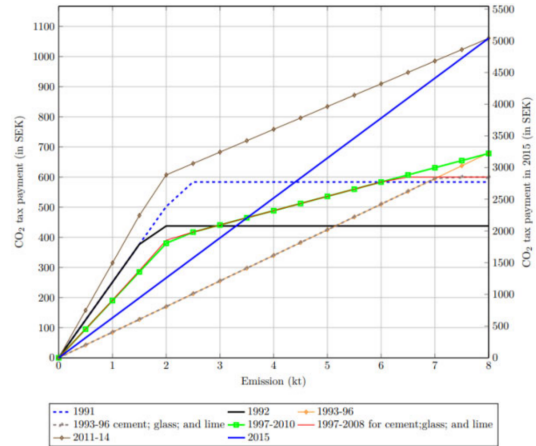


Figure 3
Changes to the carbon tax: Emissions and carbon tax payments by regime

This figure compares the carbon tax payments under the different regimes through a representative manufacturing firm. The hypothetical firm earns 50,000 SEK each year, and assumed to burn only coal in 1991 and 1992. All carbon tax payments with the exception of 2015 are shown on the vertical axis on the left side. Carbon tax payments in 2015 are shown on the vertical axis on the right side.

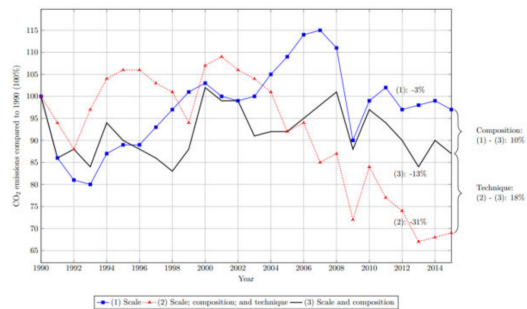


Figure 5
Carbon dioxide emissions from Swedish manufacturing (1990–2015)

This figure displays the decomposition of the Swedish carbon dioxide emission reduction. *Scale* captures how emissions would have evolved without tangible technological progress and structural changes in the manufacturing sector. *Composition* refers to the change in industry composition, and *Technique* captures the technological progress in the industrial sector.

5.8 Figure 5. Decomposition of manufacturing emissions

Read: Swedish manufacturing CO₂ emissions fell by 31% from 1990 to 2015. The decomposition attributes about 3 percentage points to scale, 10 percentage points to composition, and 18 percentage points to technique.

Economic message. Most of the aggregate reduction is not explained by lower output. The technique component is where carbon pricing is most likely to matter.

5.9 Table 3. Short-term tax-regime changes

Read: Around the 1991 and 1993 events, exemptions create large changes in marginal costs across firms. Around the 1997 event, exempt firms face a marginal cost reduction while nonexempt firms face an increase. Emission-intensity changes move in the predicted direction.

Economic message. Even short-run event comparisons support the view that firms respond to marginal carbon prices, although the paper's main estimates focus on longer-run adjustment.

Table 3

Firms with and without exemptions around the 1991, 1993, and 1997 tax changes

	Exemption (1)	No exemption (2)	Diff in groups (3)	w Ind. F.E. (4)
<i>A. 1991 and 1993 events: Marginal cost of CO₂ (SEK/Kg)</i>				
Period 1: 1990	0.0000	0.0000	0.0000 (1.0000)	
Period 2: 1991–1992	0.0000	0.2034	–0.2034 (0.0168)	
Period 3: 1993–1996	0.0842	0.0844	–0.0001 (0.0006)	
Difference periods: 2-1	0.0000 (1.0000)	0.2034 (0.0096)	–0.2034 (0.0242)	–0.2036 (0.0239)
Difference periods: 3-2	0.0842 (0.0009)	–0.1191 (0.0047)	0.2033 (0.0118)	0.2029 (0.0118)
<i>B. 1991 and 1993 events: Emissions-to-sales</i>				
Period 1: 1990	0.0865	0.0106	0.0759 (0.0053)	
Period 2: 1991–1992	0.1027	0.0110	0.0917 (0.0032)	
Period 3: 1993–1996	0.1005	0.0162	0.0843 (0.0032)	
Difference periods: 2-1	0.0162 (0.0115)	0.0004 (0.0015)	0.0158 (0.0060)	0.0165 (0.0057)
Difference periods: 3-2	–0.0022 (0.0084)	–0.0052 (0.0015)	–0.0074 (0.0050)	–0.0071 (0.0047)
<i>C. 1997 event: Marginal cost of CO₂ (SEK/Kg)</i>				
Period 1: 1993–1996	0.0844	0.0845	–0.0001 (0.0008)	
Period 2: 1997–2000	0.0756	0.1721	–0.0964 (0.0068)	
Difference in periods	–0.0087 (0.0074)	0.0876 (0.0031)	–0.0964 (0.0068)	–0.0968 (0.0065)
<i>D. 1997 event: Emissions-to-sales</i>				
Period 1: 1993–1996	0.0706	0.0170	0.0536 (0.0036)	
Period 2: 1997–2000	0.0784	0.0151	0.0633 (0.0038)	
Difference in periods	0.0078 (0.0077)	–0.0019 (0.0014)	0.0098 (0.0052)	0.0100 (0.0047)

This table reports the change in marginal cost and emission intensity for firms with (column 1) and without exemptions (column 2) around the 1991 introduction of the carbon tax and subsequent change in 1993 (panels A and B) and the 1997 event (panels C and D). Column 4 reports the difference in difference controlling for four-digit industry fixed effects. The sample includes firms from decile 10 sectors in panels A and D and is restricted to firms with observations each year over 1993–2000 in panels C and D. Panels A and C present the marginal cost of emitting CO₂ for the manufacturing firms and panels B and D the emission intensities. Standard errors are displayed in parentheses.

Table 4

Carbon pricing and firm-level carbon emission intensity

	(1) All	(2) All	(3) All	(4) All	(5) All	(6) D1-D4	(7) D5-D8	(8) D9-D10
$\Delta \ln(1 - C_{i,t-1})$	0.751*** (0.146)	0.968*** (0.157)	1.170*** (0.229)	1.115*** (0.248)	1.171*** (0.276)	2.828*** (0.884)	1.246*** (0.441)	0.838*** (0.284)
$\Delta \ln(1 - C_{i,t-2})$			0.402** (0.179)	0.585*** (0.210)	0.732*** (0.231)	1.711*** (0.656)	0.741* (0.440)	0.484* (0.258)
$\Delta \ln(1 - C_{i,t-3})$				0.377** (0.159)	0.289* (0.153)	2.184*** (0.482)	0.747** (0.349)	–0.025 (0.168)
$\sum \Delta \ln(1 - C)$	0.751*** (0.000)	0.968*** (0.000)	1.572*** (0.000)	2.077*** (0.000)	2.193*** (0.000)	6.722*** (0.002)	2.733** (0.013)	1.297*** (0.007)
Firm fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Industry-year fixed effects	No	No	No	No	Yes	No	No	No
Observations	24,943	24,757	19,485	15,001	13,948	5,529	6,284	3,130
Within R ²	.007	.014	.019	.017	.020	.076	.016	.015

This table reports OLS estimates of Equation (1). $\Delta \ln(EY)_{i,t}$ is the dependent variable. E is firm-level CO₂ emissions in kilograms (kg) and Y is firm-level, PPI-adjusted sales in Swedish krona (SEK). The sample comprises manufacturing firms in Sweden with both CO₂ emissions and sales data and with at least four consecutive observations over 1990–2015. D1–D4 include firms from the four-digit industries with emissions to sales in 1990 in the lowest 40%. D5–D8 from four-digit industries from the 5th to the 8th decile in terms of emissions intensity, and D9–D10 include firms from the highest 20% (i.e., the two highest deciles). Regressions in columns 2–8 include firm fixed effects. Regressions in columns 1–4 and 6–8 include year fixed effects and column 5 includes industry (four-digit NACE) by year fixed effects. C is the emissions cost share relative to sales for firm i in year t . The standard errors are clustered at the firm level. $\sum \Delta \ln(1 - C)$ present an F-test of joint significance. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.10 Table 4. Baseline carbon-pricing elasticities

Read: The three-lag baseline estimate gives $\sum_s \beta_s = 2.077$. With industry-year fixed effects, the estimate is 2.193. Elasticities are much larger in low-emission-intensity sectors and lower in the highest-emission sectors.

Economic message. Carbon pricing has a strong average effect, but high emitters respond less. This is important because high-emitting sectors account for most aggregate emissions.

5.11 Table 5. Robustness: EU ETS, firm size, and capital intensity

Read: The estimated elasticity remains large after accounting for EU ETS interactions, corporate-income-tax shares, labor-tax shares, employment, and capital intensity.

Economic message. The main elasticity is not driven by obvious confounds from other taxes, EU ETS participation, scale, or capital intensity.

Table 5
Carbon pricing and carbon emission intensity: EU ETS, firm size, and capital intensity

	(1) All	(2) D9-D10	(3) CIT	(4) Labor	(5) All	(6) All	(7) All
$\Delta \ln(1 - C)_{(i,t-1)}$	1.414*** (0.327)	1.025*** (0.392)	1.115*** (0.247)	1.134*** (0.265)	1.008*** (0.253)	1.012*** (0.252)	1.011*** (0.253)
$\Delta \ln(1 - C)_{(i,t-2)}$	0.684*** (0.246)	0.508 (0.329)	0.586*** (0.210)	0.610*** (0.224)	0.512** (0.205)	0.514** (0.205)	0.515** (0.205)
$\Delta \ln(1 - C)_{(i,t-3)}$	0.351* (0.188)	-0.208 (0.218)	0.374** (0.160)	0.387** (0.169)	0.245* (0.137)	0.243* (0.137)	0.245* (0.138)
EU ETS	0.000 (0.001)	0.000 (0.001)					
$x \Delta \ln(1 - C)_{(i,t-1)}$	-0.953** (0.390)	-0.470 (0.407)					
$x \Delta \ln(1 - C)_{(i,t-2)}$	-0.317 (0.308)	0.008 (0.392)					
$x \Delta \ln(1 - C)_{(i,t-3)}$	0.180 (0.303)	0.630* (0.378)					
$\Delta \ln(1 - T)_{(i,t-1)}$			0.004 (0.010)	0.006 (0.004)			
$\Delta \ln(1 - T)_{(i,t-2)}$			-0.011 (0.009)	0.006 (0.006)			
$\Delta \ln(1 - T)_{(i,t-3)}$			-0.004 (0.006)	0.005 (0.004)			
$\ln(EMP)_{(i,t)}$					-0.002*** (0.001)		-0.002*** (0.001)
$\ln(CAP/EMP)_{(i,t)}$						0.001*** (0.000)	0.001*** (0.000)
$\sum \Delta \ln(1 - C)$	1.359** (0.017)	1.493** (0.030)	2.075*** (0.001)	2.131*** (0.000)	1.765*** (0.000)	1.769*** (0.000)	1.770*** (0.000)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,001	3,130	15,001	13,952	14,828	14,789	14,789
Within R ²	.021	.019	.018	.018	.027	.023	.030

Table 5 reports OLS estimates of Equation (1). $\Delta \ln(EY)_{i,t}$ is the dependent variable. E is firm-level CO₂ emissions in kilograms (kg) and Y is firm-level, PPI-adjusted sales in Swedish krona (SEK). The sample comprises manufacturing firms in Sweden with both CO₂ emissions and sales data and with at least four consecutive observations over 1990–2015. In column 2 (D9 and 10), we only include firms from the four-digit industries with emissions to sales in 1990 in the highest 20% (i.e., the two highest deciles). All regressions include firm and year fixed effects. C is the emissions cost share relative to sales for firm i in year t . T is the share of corporate income taxes (CIT) relative to sales for firm i in year t in column 3 and the share of labor taxes (Labor) to sales for firm i in year t in column 4. Labor tax data ends in 2013 so the sample in column 4 is 1990–2013. *EU ETS* is an indicator variable taking on the value one when a firm-year has at least one plant regulated under EU ETS. The standard errors are clustered at the firm level. $\sum \Delta \ln(1 - C)$ present an F-test of joint significance. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Carbon pricing and carbon emission intensity: PACE and mobility

	(1) PACE	(2) High	(3) Low PACE	(4) High mobility	(5) Low PACE	(6) High mobility
$\Delta \ln(1 - C)_{(i,t-1)}$	1.320*** (0.394)	1.088*** (0.297)	1.375** (0.586)	1.288*** (0.491)	0.942*** (0.335)	1.685*** (0.651)
$\Delta \ln(1 - C)_{(i,t-2)}$	0.849*** (0.298)	0.527** (0.261)	1.100*** (0.346)	0.614 (0.425)	0.552* (0.296)	0.368 (0.533)
$\Delta \ln(1 - C)_{(i,t-3)}$	0.832*** (0.213)	0.281 (0.202)	0.304 (0.279)	1.027*** (0.267)	0.228 (0.199)	0.399 (0.598)
$\sum \Delta \ln(1 - C)$	3.000*** (0.000)	1.895*** (0.001)	2.779*** (0.006)	2.928*** (0.003)	1.721*** (0.006)	2.452* (0.059)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,671	7,568	3,023	3,591	4,773	2,673
Within R ²	.024	.016	.034	.023	.013	.035

This table reports OLS estimates of Equation (1). $\Delta \ln(EY)_{i,t}$ is the dependent variable. E is firm-level CO₂ emissions in kilograms (kg) and Y is firm-level, PPI-adjusted sales in Swedish krona (SEK). The sample comprises manufacturing firms in Sweden with both CO₂ emissions and sales data and with at least four consecutive observations over 1990–2015. All regressions include firm and year fixed effects. C is the emissions cost share relative to sales for firm i in year t . The standard errors are clustered at the firm level. $\sum \Delta \ln(1 - C)$ present an F-test of joint significance. * $p < .1$; ** $p < .05$; *** $p < .01$.

Results are shown in columns 3–6 of Table 6. Firms in sectors defined as low

5.12 Table 6. PACE and mobility

Read: Low-PACE sectors have an elasticity around 3.0, while high-PACE sectors have an elasticity around 1.9. High-PACE, low-mobility sectors have an elasticity around 1.7; high-PACE, high-mobility sectors have a higher but less precisely estimated elasticity around 2.45.

Economic message. Lower abatement costs produce stronger responses. Mobility matters most when abatement costs are high, because firms may have more scope to shift production or strategy.

5.13 Table 7. Carbon leakage classifications

Read: Firms not on the leakage list have an elasticity around 2.6, while firms on the leakage list have an elasticity around 1.9. Trade-only leakage-list sectors show a larger elasticity, while

emission-intensive leakage-list sectors show a lower elasticity around 1.5.

Economic message. The sectors most exposed to carbon leakage and high emissions are also the sectors where carbon pricing produces weaker measured reductions in emission intensity.

Table 7
Carbon pricing and carbon emission intensity: Carbon leakage

	(1)	(2)	(3)	(4)
	Leakage list		Leakage list Yes	
	No	Yes	Trade only	Emission
$\Delta \ln(1 - C)_{(i,t-1)}$	1.257*** (0.457)	1.062*** (0.301)	1.605** (0.645)	0.956*** (0.329)
$\Delta \ln(1 - C)_{(i,t-2)}$	0.565 (0.419)	0.609** (0.247)	0.793 (0.504)	0.644** (0.303)
$\Delta \ln(1 - C)_{(i,t-3)}$	0.764** (0.332)	0.214 (0.170)	0.950*** (0.308)	−0.057 (0.193)
$\sum \Delta \ln(1 - C)$	2.585*** (0.009)	1.885*** (0.000)	3.348*** (0.008)	1.543*** (0.007)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	7,228	7,737	5,805	1,932
Within R ²	.015	.021	.039	.021

This table reports OLS estimates of Equation (1). $\Delta \ln(E/Y)_{i,t}$ is the dependent variable. E is firm-level CO₂ emissions in kilograms (kg) and Y is firm-level, PPI-adjusted sales in Swedish krona (SEK). The sample comprises manufacturing firms in Sweden with both CO₂ emissions and sales data and with at least four consecutive observations over 1990–2015. All regressions include firm and year fixed effects. C is the emissions cost share relative to sales for firm i in year t . The standard errors are clustered at the firm level. $\sum \Delta \ln(1 - C)$ present an F-test of joint significance. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Carbon pricing and carbon emission intensity: Financing constraints

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	All sectors		D1-D4 sectors		D9-D10 sectors		Low PACE		High PACE	
A. Publicly listed										
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
$\sum \Delta \ln(1 - C)$	2.220*** (0.001)	0.959 (0.206)	7.478*** (0.002)	3.902*** (0.000)	2.323*** (0.004)	0.401 (0.628)	3.173 (0.277)	2.659*** (0.019)	2.525*** (0.001)	0.739 (0.408)
Observations	2,107	6,535	595	2,531	736	1,307	464	2,083	1,567	4,207
B. Large firm										
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
$\sum \Delta \ln(1 - C)$	2.115*** (0.009)	0.585 (0.296)	4.100*** (0.000)	4.607*** (0.000)	1.948* (0.076)	0.680 (0.266)	1.920* (0.097)	3.811*** (0.031)	2.066** (0.027)	0.494 (0.446)
Observations	4,138	4,504	1,581	1,545	886	1,157	1,254	1,293	2,801	2,973
C. High-dividend payer										
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
$\sum \Delta \ln(1 - C)$	2.699*** (0.000)	0.743 (0.301)	4.053*** (0.015)	4.243*** (0.000)	3.641*** (0.000)	-0.024 (0.971)	2.872 (0.113)	2.441*** (0.042)	2.671*** (0.002)	0.659 (0.450)
Observations	4,209	4,433	1,558	1,568	930	1,113	1,273	1,274	2,822	2,952
D. Mature firm										
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
$\sum \Delta \ln(1 - C)$	2.934*** (0.000)	0.562 (0.450)	4.457*** (0.000)	3.907*** (0.000)	3.072*** (0.000)	0.184 (0.793)	4.654*** (0.004)	1.615 (0.144)	2.838*** (0.002)	0.435 (0.632)
Observations	3,814	4,779	1,489	1,613	799	1,232	1,167	1,365	2,549	3,194

This table reports $\sum \Delta \ln(1 - C)$ and the corresponding F-test based on OLS estimates of Equation (1) where $\Delta \ln(E/Y)_{i,t}$ is the dependent variable. See Tables B.6, B.7, B.8, and B.9 in the Internet Appendix for detailed regression results for each panel. The sample period is 1996–2015. All regressions include firm and year fixed effects. *Public (Private) firm* is an indicator variable taking on the value one (zero) if the firm is (not) listed on a Swedish stock exchange. A firm is considered publicly listed if at least one firm in the corporate group is listed at least once during the sample period. *Large (Small) firm* is an indicator variable taking on the value one (zero) if the firm is above (below) the median in book value of total assets (averaged over the sample period and measured at the corporate group level) within its four-digit NACE industry. *High (Low) dividend firm* is an indicator variable taking on the value one (zero) if the firm is above (below) the median in dividend payout divided by book value of total assets (averaged over the sample period and measured at the corporate group level) within its four-digit NACE industry. *Mature (Young) firm* is an indicator variable taking on the value one (zero) if the firm's founding year (measured at the corporate group level) is below (above) the median within its four-digit NACE industry. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.14 Table 8. Financing constraints

Read: Public, large, high-dividend, and mature firms have stronger estimated elasticities than private, small, low-dividend, and young firms. The gap is especially visible in high-PACE sectors.

Economic message. Carbon pricing does not work only through incentives. Firms also need the financial capacity to invest in abatement technologies or reorganize production.

5.15 Table 9. Aggregate economic magnitude

Read: Using the baseline elasticity, 2015 emissions intensity would be 47% higher without carbon pricing. After accounting for PACE and mobility heterogeneity, aggregate emissions intensity would be about 30% higher. Financial-constraint splits imply even larger differences between constrained and unconstrained firms.

Economic message. The estimated firm-level elasticities are economically large enough to explain a substantial share of Sweden's manufacturing emissions decline.

Table 9
Economic magnitude based on the 2015 carbon pricing change and emissions intensities

	(1) Share CO ₂	(2) Elasticity	(3) CO ₂ intensity	(4) Without tax	(5) Relative
<i>A. PACE, mobility, and aggregate emissions</i>					
All	1.0000	2.0769	0.0049	0.0071	47%
Low pace & Low mobility	0.0415	2.7789	0.0033	0.0057	74%
Low pace & High mobility	0.0125	2.9284	0.0025	0.0042	68%
High pace & Low mobility	0.9021	1.7213	0.0077	0.0098	27%
High pace & High mobility	0.0438	2.4516	0.0049	0.0068	38%
Aggregate emissions					30%
<i>B. PACE, Leakage list, and deciles</i>					
Low pace sectors	0.0541	3.0003	0.0029	0.0054	83%
High pace sectors	0.9459	1.8948	0.0067	0.0087	31%
Not on leakage list	0.0758	2.5853	0.0039	0.0060	52%
On leakage list	0.9242	1.8850	0.0058	0.0078	33%
Deciles 1-4	0.0310	6.7230	0.0025	0.0069	175%
Deciles 5-8	0.0591	2.7340	0.0039	0.0069	78%
Deciles 9-10	0.9099	1.2970	0.0142	0.0174	23%
<i>C. Ownership, size, dividend payout, and age</i>					
Public firm	0.4684	2.2195	0.0074	0.0103	39%
Private firm	0.5316	0.9591	0.0044	0.0050	14%
Large firm	0.7077	2.1150	0.0047	0.0065	38%
Small firm	0.2923	0.5854	0.0049	0.0056	12%
High dividend firm	0.4110	2.6990	0.0047	0.0071	51%
Low dividend firm	0.5890	0.7429	0.0050	0.0050	0%
Mature firm	0.6616	2.9335	0.0045	0.0076	69%
Young firm	0.3384	0.5620	0.0051	0.0057	13%

This table reports the share of aggregate CO₂ emissions across subsamples in 2015 (in column 1), the estimated elasticity (in column 2), the actual CO₂-to-sales in 2015 for each subsample (in column 3), the value from subtracting the product of the elasticity and actual carbon pricing change in 2015 to the actual CO₂-to-sales (in column 4), and the ratio of column 4 and column 3 (in column 5).

6 Short summary

- Sweden's carbon tax created long-run variation in marginal carbon prices across manufacturing firms.
- The paper estimates firm-level carbon-pricing elasticities using emission intensity and marginal emissions cost shares.
- The baseline three-year elasticity is about **2.1**; adding industry-year fixed effects gives about **2.2**.
- High-emitting sectors respond less than low-emitting sectors.
- Low-PACE sectors respond more than high-PACE sectors.
- Leakage-list and emission-intensive sectors have lower elasticities, consistent with higher abatement costs and leakage concerns.
- Less financially constrained firms respond much more strongly, especially in high-abatement-cost sectors.
- A counterfactual calibration suggests that 2015 Swedish manufacturing emissions intensity would have been about **30% higher** without carbon pricing.

7 Conclusion

7.1 Core conclusion

The paper provides strong firm-level evidence that carbon pricing reduces manufacturing emission intensity. Its main contribution is not just showing a treatment effect from introducing a carbon tax, but estimating a long-run elasticity using many years of marginal price variation.

7.2 Economic intuition

A carbon price changes the payoff from reducing emissions, but the response depends on whether abatement is technologically feasible and financially affordable. Firms with low abatement costs or better financing access reduce emissions more. High-emitting, high-PACE, and financially constrained firms respond less.

7.3 Takeaway

Carbon taxes can be powerful, but policy design matters. Exemptions that lower marginal prices weaken incentives for the firms that often emit the most. Complementary financing or technology policies may raise the effectiveness of carbon pricing where abatement is costly.